

Conversion

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Decimal to Binary

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Binary to Decimal

Decimal to Octal

Octal to Decimal

Decimal to Hexadecimal Hexadecimal to Decimal

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Binary to Octal

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Octal to Binary

Binary to Hexadecimal

Hexadecimal to Binary

Decimal To Binary

- ***Repeated-division method (For Integer No)***
- The number is successively divided by 2 and its remainders recorded, and finally assembling all the remainders from bottom to top.
- Means last (bottom) remainder being the most significant bit (MSB) and first (top) remainder being the least significant bit (LSB).

Decimal to Binary

- $(43)_{10} = (\quad)_2$

2	43
2	21
2	10
2	5
2	2
2	1
	0

1 • LSB



- Write in this order

1 • MSB

- $(43)_{10} = (101011)_2$

Decimal to Binary

- For *fractional number* the procedure is to successively multiply the decimal fraction by the 2 and collect all members to the left of the decimal point
- finally assembling all the remainders from top to bottom.

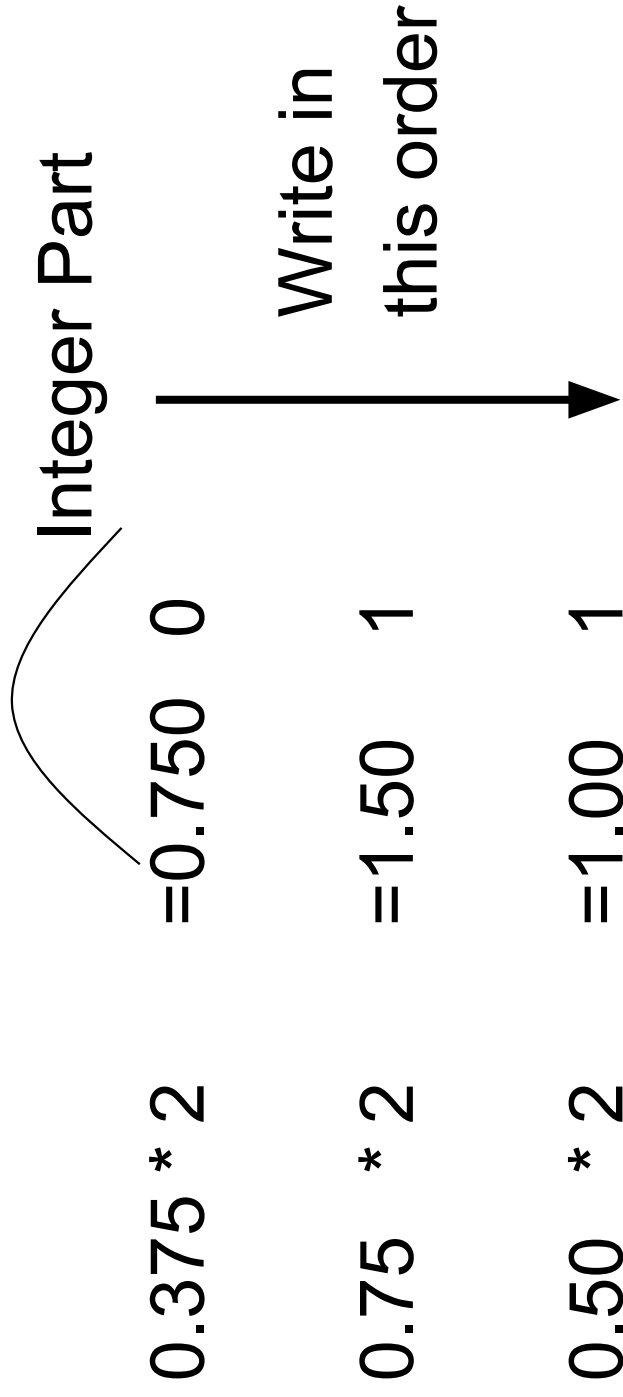
Decimal To Binary

- $(0.375)_{10} = (\quad)_2$

$0.375 * 2$	$= 0.750$	0
$0.75 * 2$	$= 1.50$	1
$0.50 * 2$	$= 1.00$	1

Integer Part

Write in
this order



- $(0.375)_{10} = (. 0 1 1)_2$

Decimal to Octal

- ***Repeated-division method (For Integer No)***
- The number is successively divided by 8 and its remainders recorded, and finally assembling all the remainders from bottom to top.
- Means last (bottom) remainder being the most significant bit (MSB) and first (top) remainder being the least significant bit (LSB).

Decimal to Octal

- $(266)_{10} = (\quad)_8$

8	266
8	33
8	4
0	

2

• LSB

• Write in
this order

1

4

• MSB

- $(266)_{10} = (412)_8$

Decimal to Octal

- For *fractional number* the procedure is to successively multiply the decimal fraction by the 8 and collect all members to the left of the decimal point
- finally assembling all the remainders from top to bottom.

Decimal To Octal

$$\bullet (0.375)_{10} = (\quad)_8$$

$$0.375 * 8 = 3.000 \quad 3$$

Integer Part

$$\bullet (0.375)_{10} = (0.3)_8$$

$$\bullet (0.015625)_{10} = (\quad)_8$$

$$0.015625 * 8 = 0.125 \quad 0$$

Integer Part

Write in $\downarrow$ this order

$$0.125 * 8 = 1.000 \quad 1$$

$$\bullet (0.015625)_{10} = (0.01)_8$$

Decimal To Hexadecimal

- ***Repeated-division method (For Integer No)***
- The number is successively divided by 16 and its remainders recorded, and finally assembling all the remainders from bottom to top.
- Means last (bottom) remainder being the most significant bit (MSB) and first (top) remainder being the least significant bit (LSB).

Decimal To Hexadecimal

- $(423)_{10} = (\quad)_{16}$

16	423
16	26
16	1
0	

7 • LSB • Write in
A ↑ this order
1 • MSB

- $(423)_{10} = (1A7)_{16}$

Decimal To Hexadecimal

- For *fractional number* the procedure is to successively multiply the decimal fraction by the 16 and collect all members to the left of the decimal point
- finally assembling all the remainders from top to bottom.

Decimal To Hexadecimal

- $(0.03125)_{10} = (\quad)_{16}$

$0.03125 * 16$	$= 0.5$	0
$0.5 * 16$	$= 8.0$	8

Integer Part

↓

- $(0.03125)_{10} = (0.08)_{16}$

2

- Binary to Decimal
- Octal to Decimal
- Hexadecimal to Decimal

Binary to Decimal

- Binary number can be converted to its decimal equivalent simply by summing together the weights of the various position in the binary number which contains a 1. (For Integer No)

Binary to Decimal

$$(11011)_2 = (\quad)_{10} \quad \text{Position of each bit}$$

4	3	2	1	0
1	1	0	1	1

$$= 1 * 2^4 + 1 * 2^3 + 0 * 2^2 + 1 * 2^1 + 1 * 2^0$$

$$= 16 + 8 + 2 + 1$$

Positional weight
for each bit

$$= (27)_{10}$$

Binary to Decimal

- Converting Binary Fractions to Decimal

Binary to Decimal

$$(0.0101)_2 = (\quad)_{10}$$

Positional weight
for each bits

$$2^{-1} \quad 2^{-2} \quad 2^{-3} \quad 2^{-4}$$

0	1	0	1
---	---	---	---

$$= 0 * 2^{-1} + 1 * 2^{-2} + 0 * 2^{-3} + 1 * 2^{-4}$$

$$= 0 + 0.25 + 0 + 0.0625$$

$$= (0.3125)_{10}$$

Octal to Decimal

- Octal number can be converted to its decimal equivalent simply by the sum of the products of each digit value and its positional value

Octal to Decimal

$$(372)_8 = (\quad)_{10}$$

2 1 0

3	7	2
---	---	---

$$= 3 * 8^2 + 7 * 8^1 + 2 * 8^0$$

$$= 3 * 64 + 7 * 8 + 2 * 1$$

$$= (250)_{10}$$

Octal to Decimal

$$(24.6)_8 = (\quad)_{10} \text{ (For Fractional Part)}$$

$$1 \qquad 0 \qquad -1$$

2	4	6
---	---	---

$$= 2 * 8^1 + 4 * 8^0 + 6 * 8^{-1}$$

$$= (20.75)_{10}$$

Hexadecimal to Decimal

- Hexadecimal number can be converted to its decimal equivalent simply by the sum of the products of each digit value and its positional value

Hexadecimal to Decimal

$$(356)_8 = (\quad)_{10}$$

2 1 0

3	5	6
---	---	---

$$= 3 * 16^2 + 5 * 16^1 + 6 * 16^0$$

$$= 768 + 80 + 6$$

$$= (854)_{10}$$

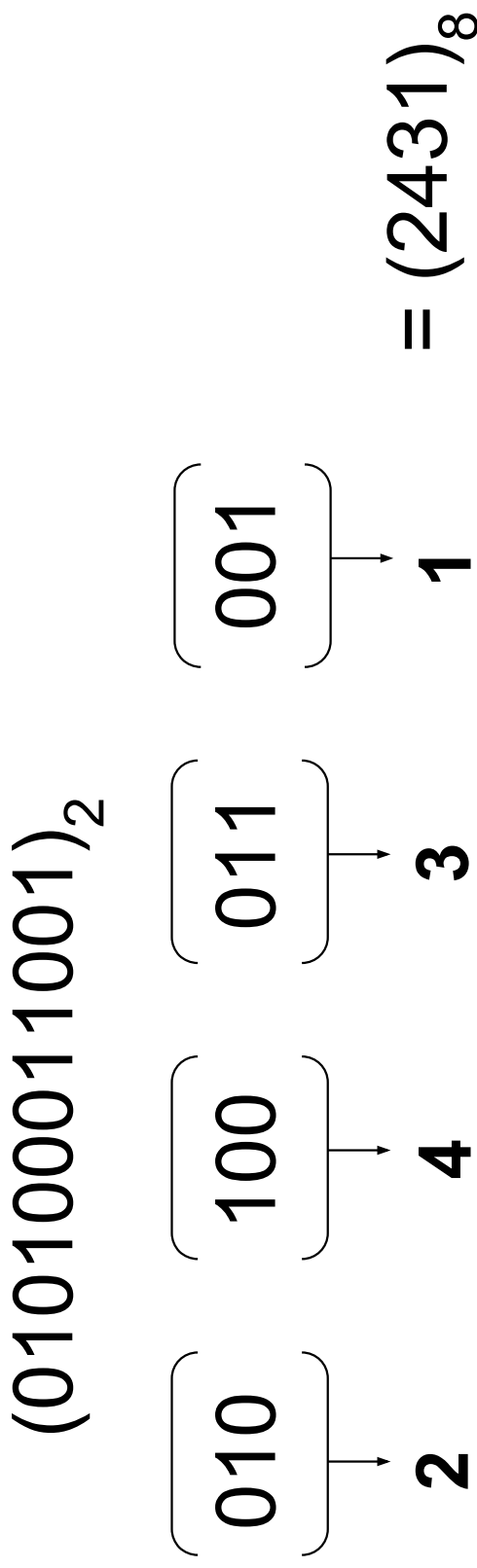
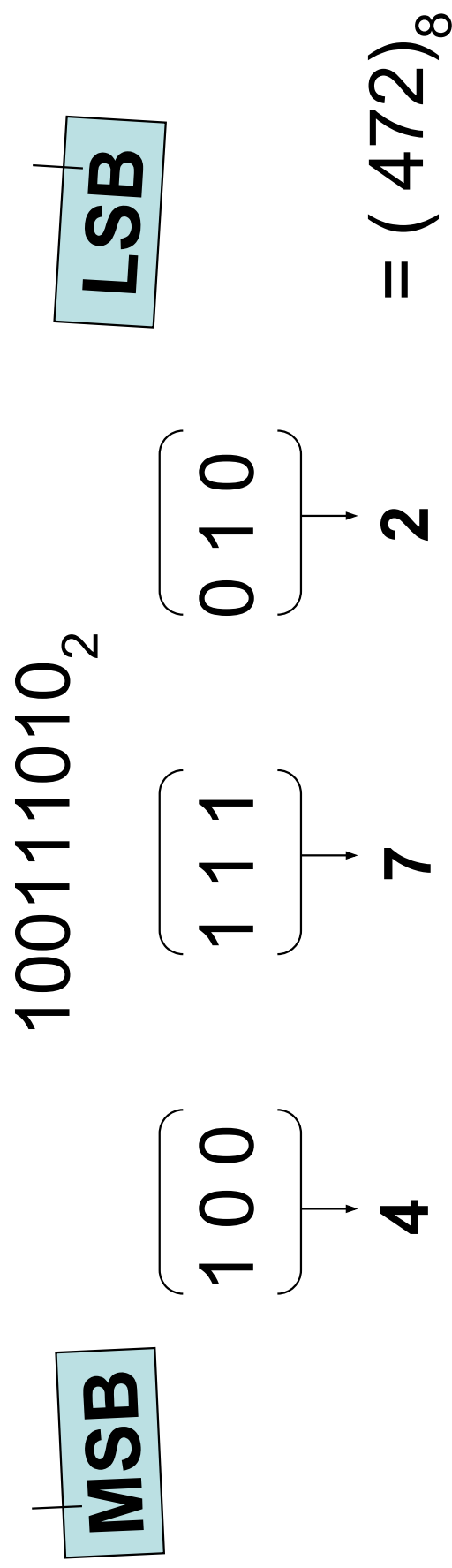
3

- Binary to Octal
- Binary to Hexadecimal

Binary to Octal

- The bits of the binary integer are grouped into groups of ***THREE*** bits starting from the LSB.
- Then each group is converted to its octal equivalent.

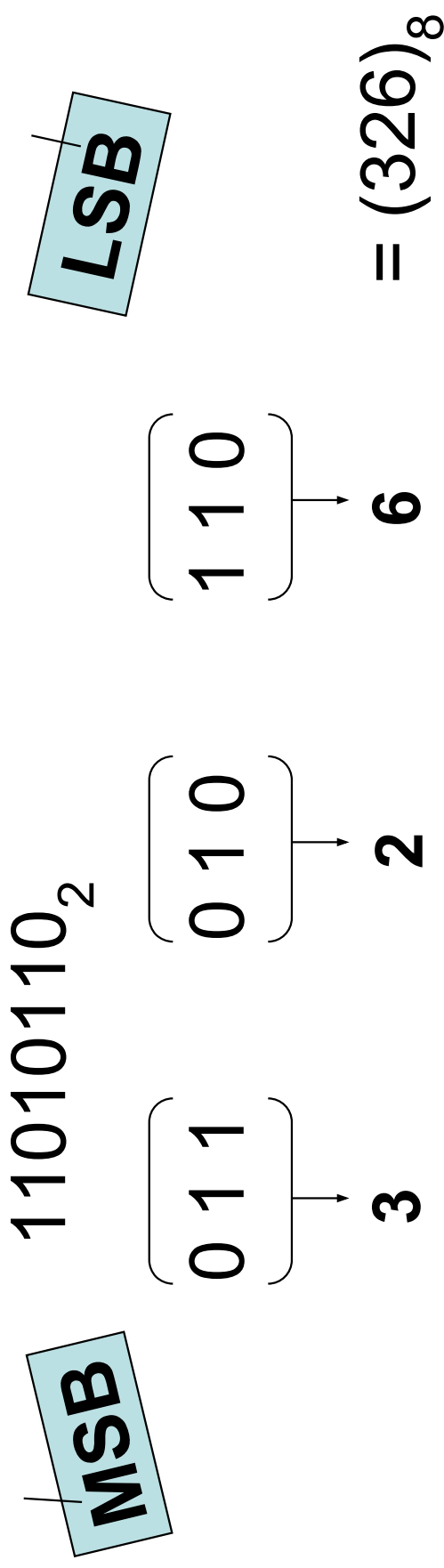
Binary to Octal



Binary to Octal

- Q. What happened ? If the binary number will not have even groups of 3 bits.
- A. For those case we can add one or two 0s to the left of MSB of the binary number to fill out the last group.

Binary to Octal



O added here to
make it group of
3 bits

Binary to Octal

- For Fractional part the same process will apply
- But after the binary point, the groups of 3 bits are made starting from left – to – right.
- If any bits are missing from a groups of bits so we can add 0 from the LSB side.

Binary to Octal

.0101₂

0 1 0



2

1 0 0



4

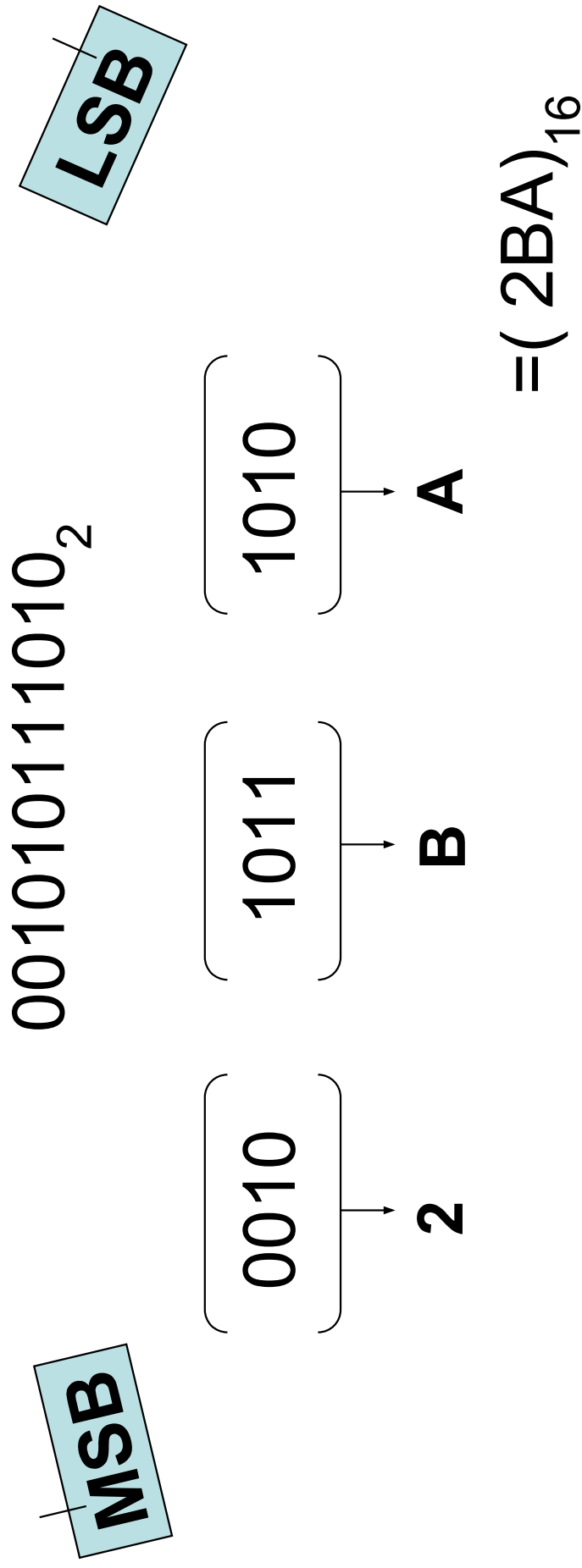
0 added here to
make it group of
3 bits

= (.24)₈

Binary to Hexadecimal

- The bits of the binary integer are grouped into groups of ***FOUR*** bits starting from the LSB.
- Then each group is converted to its Hexadecimal equivalent.

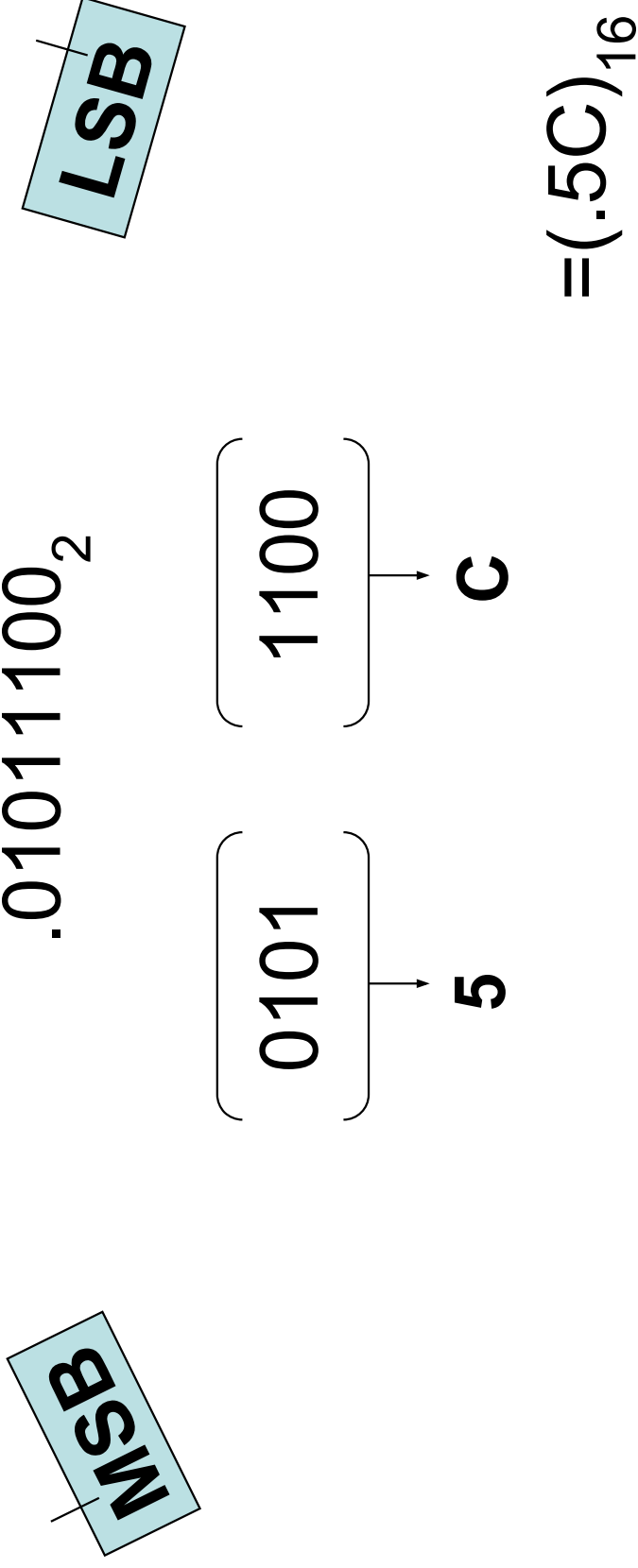
Binary to Hexadecimal



Binary to Hexadecimal

- For Fractional part
- after the binary point, the groups of 4 bits are made starting from left – to – right.
- If any bits are missing from a groups of bits so we can add 0 from the LSB side.

Binary to Hexadecimal



Question

Binary to Decimal

1. 10010
2. 101010
3. 1010100.011
4. 10110.01
5. 101011111

Binary to Octal

1. 11111011110101
2. 1011110100011000111
3. 11110111100
4. 101011111
5. 10110111011011

Binary to Hexadecimal

1. 1010110110111
2. 1111011101011011
3. 11010111010111
4. 10011011101
5. 1111101110101111

4

- Octal to Binary
- Hexadecimal to Binary

Octal to Binary

Octal	0	1	2	3	4	5	6	7
Binary	000	001	010	011	100	101	110	111

Octal to Binary

- The conversion from octal to binary is performed by converting each digit to its 3-bit binary equivalent

Octal to Binary

$$\bullet = (472.12)_8 = (\quad)_2$$

4	7	2	.	1	2
$\overline{100}$	$\overline{111}$	$\overline{010}$	$\bullet$	$\overline{001}$	$\overline{010}$

$$= (100111010.001010)_2$$

Hexadecimal to Binary

- The conversion from Hexadecimal to binary is performed by converting each digit to its 4-bit binary equivalent

Hexadecimal to Binary

$$= (9 \text{ F } 2)_{16} = (\quad)_2$$

9

F

2

$\overbrace{1001}$

$\overbrace{1111}$

$\overbrace{0010}$

$$= (100111110010)_2$$

END